Amazon Sales Data Analysis — README

This repository contains a complete analysis of Amazon Men's Wearables sales data using Python. It includes data cleaning, preprocessing, exploratory data analysis (EDA), and insights derived from various statistical summaries and visualizations.

Project Overview

This project analyzes raw Amazon sales data and extracts meaningful business insights. It covers:

- Data loading and cleaning
- Handling missing values
- Exploring fulfilment methods
- Order status analysis
- Category-wise sales performance
- Size distribution for apparel
- Courier status performance
- Daily revenue trends
- Top shipping cities and states

The purpose is to understand buyer behaviour, product performance, logistics efficiency, and operational patterns.

Files in This Repository

File	Description
Amazon_Sale_Report.csv	Raw dataset of Amazon orders
Amazon_sales_data_analysis.ipynb	Jupyter notebook containing all analysis steps
README.md	This documentation file summarizing all work

Technologies Used

- **Python**
- **Pandas** (data cleaning & analysis)
- **NumPy** (math operations)
- **Matplotlib** (visualizations)
- **Jupyter Notebook**

Steps Performed in the Notebook

Below is a concise explanation of each major step in the project:

Importing Required Libraries

Loaded the necessary libraries for data manipulation and visualization.

Reading the Dataset

Loaded the CSV file using pd.read_csv() and stored it as a DataFrame.

Initial Data Inspection

Used:

- data.head() to preview the first few rows
- data.info() to list data types and missing values
- data.shape() to check dataset size

Cleaning the Dataset

Performed tasks like:

- Dropping unnecessary columns (index, New, PendingS)
- Converting Date column to datetime type
- Standardizing state names to lowercase

Order Status Analysis

- Extracted unique status values
- Computed counts and percentages
- Visualized distribution using bar charts

Fulfilment Analysis

- Checked fulfilment methods (Amazon vs Merchant)
- Calculated counts and percentages

Fulfilled-By Channel

- Checked unique values & null counts
- Verified Easy Ship as dominant type

Sales Channel Exploration

- Identified orders from Amazon vs Non-Amazon channels

Shipping Service Levels

- Compared Expedited vs Standard shipping usage

Category-Level Analysis

Performed grouped aggregations:

- Total quantity sold per category

- Total revenue per category
- Average selling price per item

Category + Size Deep-Dive

- Size distribution per category
- Identified most sold size
- Identified highest revenue size-category pair

Courier Status Analysis

- Computed counts and percentages for courier outcomes

Daily Revenue Trend

- Aggregated sales by date
- Identified highest and lowest revenue days
- Visualized revenue trend

Top Shipping Cities & States

- Identified top 10 cities and states by shipping volume
- Visualized both using bar charts

Key Insights

highlights from the analysis:

- **Expedited shipping dominates**, showing customers prefer faster delivery.
- **Amazon fulfilment handles majority of orders**, implying higher reliability.
- **T-Shirt category leads in quantity sold**, while other categories may lead in revenue.
- **Certain city/state clusters show very high demand**, useful for targeted inventory planning.
- **Daily revenue trends reveal peak and low-performing days** which can guide promotional strategies.

How to Run This Project

1. Install Python 3.x
2. Install required libraries:
3. pip install pandas numpy matplotlib
4. Open the Jupyter notebook:
5. jupyter notebook Amazon_sales_data_analysis.ipynb
6. Run each cell in sequential order.

Contributions

Feel free to fork the repository and submit pull requests for improvements or additional analysis modules.

Contact

For any help or clarification, open an issue in the repository or reach out to the author.
